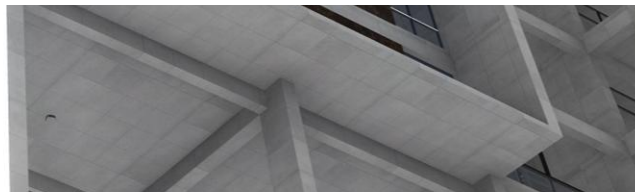
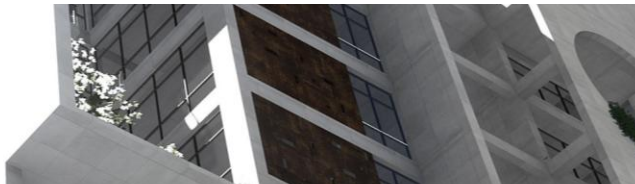
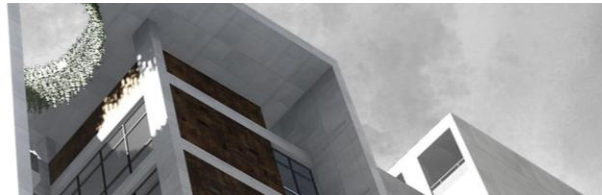




A COLLECTION OF DESIGNS AND CREATIVE WORKS BY

TASNIM NOWER ISSANA

portfolio



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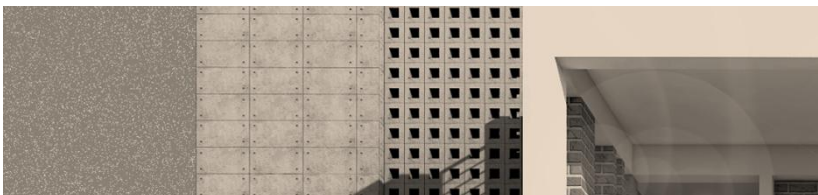
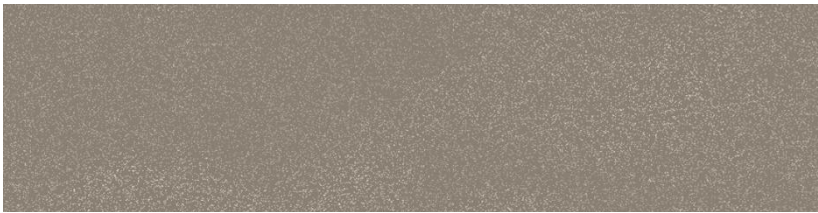
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BACHELOR OF ARCHITECTURE



P R O F E S S I O N A L W O R K

OVERVIEW

While designing an Institute building, it is imperative to imagine the project's scale for an urban public realm and design on an axis. In this design I make the pedestrian entry more welcoming. The pedestrian-friendly campus is axially planned, directing the movement and giving a sense of order and circulation. It also accommodates transitional and social activities in its open lawns. It is easy for newcomers to get lost in the maze of buildings and pathways at a University campus so that I have created a grand plaza. A focus helps visually dominate a space acting as a landmark to help people navigate. This focus also has cultural and social significance and can be a center for recreational activities. It is an integral part of the design that draws eyes towards it.

Professional Work , Year | 2024

Location: Dhaka , Bangladesh

Designation : Chief Architect

Project type : Institutional



RENDERS | PLAN



GROUND FLOOR PLAN



FIRST FLOOR PLAN

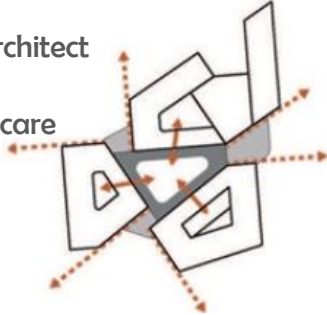
SIMEC HOSPITAL

Professional Work , Year | 2022

Location: Dhaka , Bangladesh

Designation : Chief Architect

Project type : Healthcare



OVERVIEW

This is a hospital accommodating more than 250 beds. I believe that the design and planning of a hospital play an important role in the healing process. My aim is to create a space that responds to the needs of everyone who lives, works, and functions within it.

One of the biggest challenges in hospital design is managing circulation and entry-exit points. I have addressed this by separating the entrances and exits for IPD, OPD, and Emergency services, while creating an internal lobby that acts as the heart of the hospital, guiding everyone to their destination. I have also incorporated numerous green terraces for patients, creating peaceful spaces that connect them with nature. My hope is that everyone leaves this hospital with a sense of healing and positivity in their hearts.



PLAN | RENDERS



MASTERPLAN OF SIMEC FOUNDATION

OVERVIEW

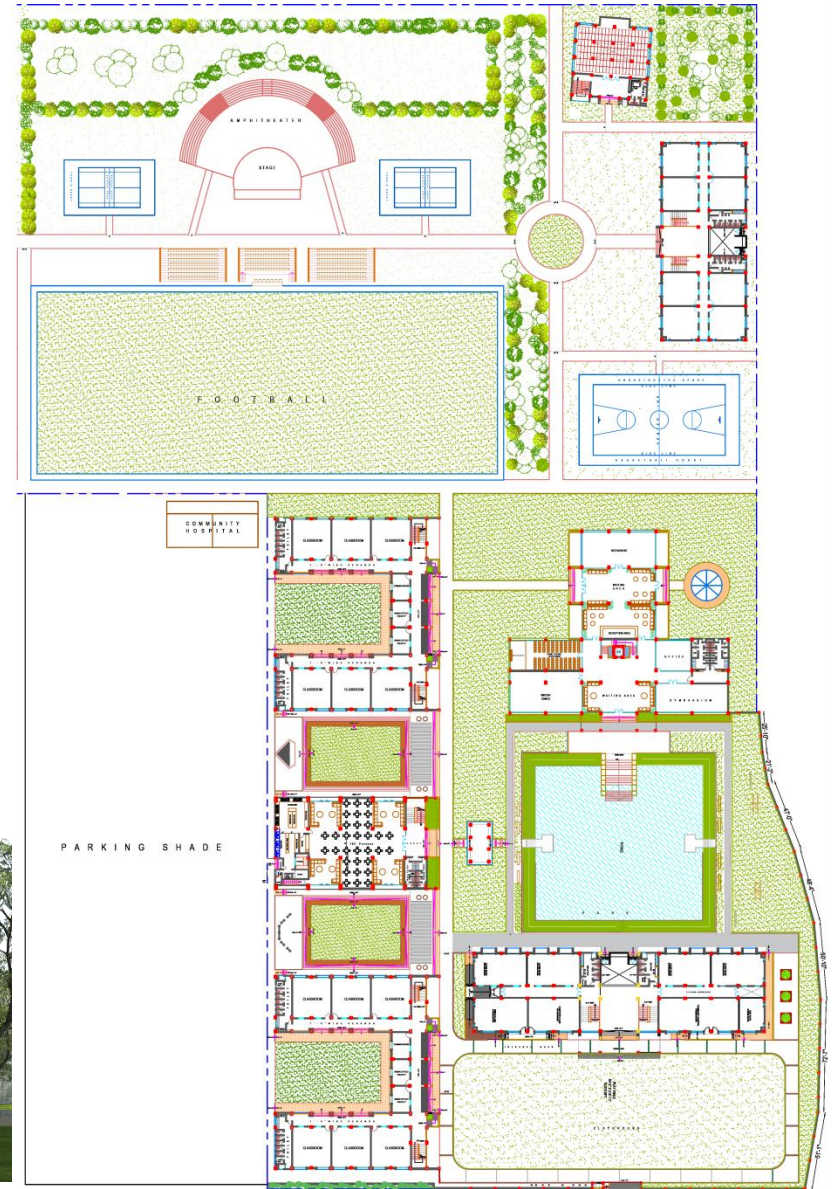
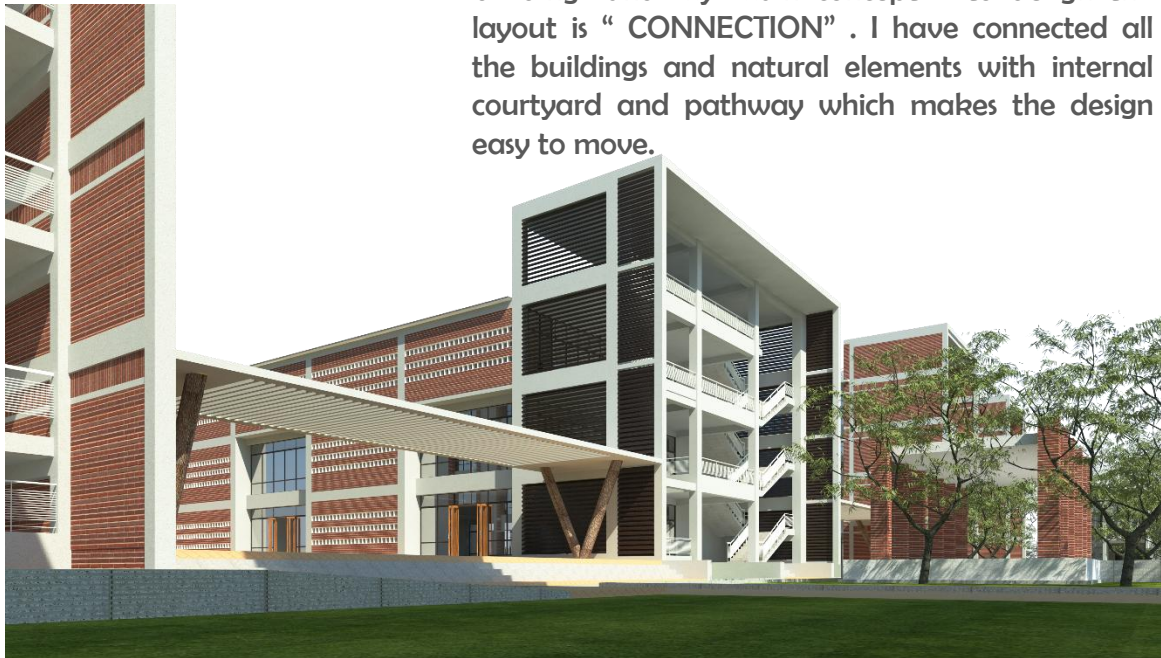
Professional Work , Year | 2021

Location: Dhaka , Bangladesh

Designation : Chief Architect

Project type : Vocational institute

It is master plan based project where the key functions are 6 institute / training building , 1 building that serves the recreational purposes such as library , restaurant , multipurpose hall , 1 dedicated research building , a mosque , amphitheatres , play zones . While designing I have faced the challenge which is the site . The site is east west oriented and there are an existing large scale pond which I have to incorporate with my layout . So after facing all the obstacles I have treated the building with sun shading device which I have made the main elevation feature of the buildings and my main concept is to design this layout is “ CONNECTION” . I have connected all the buildings and natural elements with internal courtyard and pathway which makes the design easy to move.



RENDERS



RESIDENTIAL PROJECT

Professional Work , Year | 2021

Location: Jolshiri, Dhaka , Bangladesh

Designation : Chief Architect

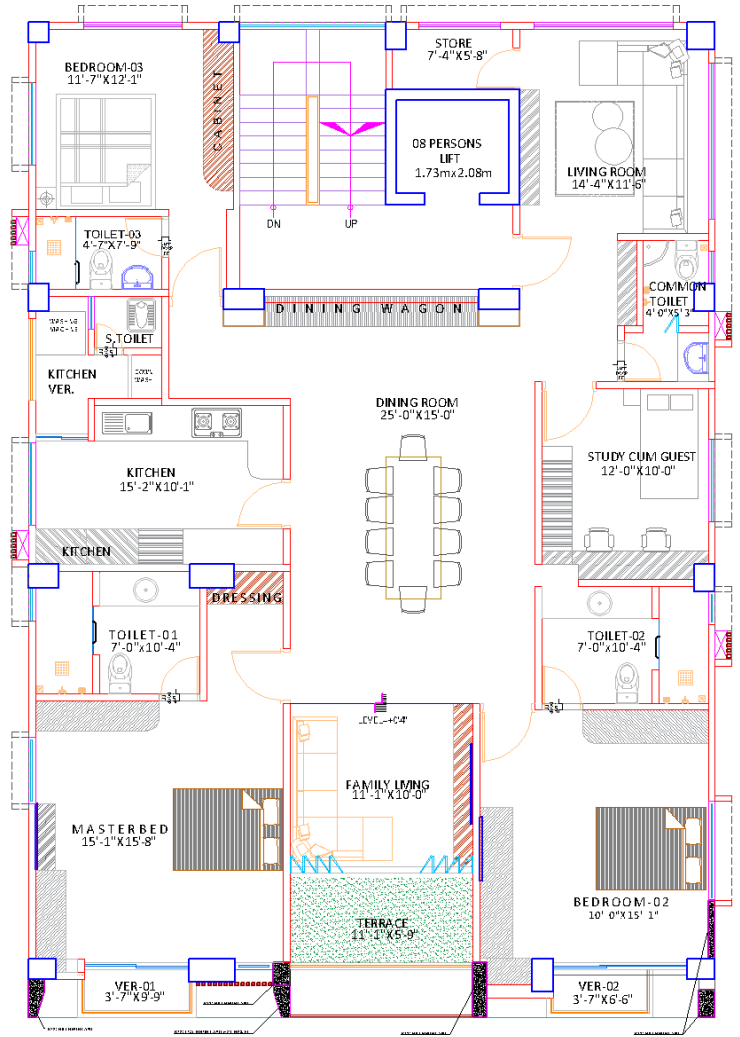
Project type : Residential

OVERVIEW

As an architect when I design apartments my focus is on blending practicality, aesthetics and eco friendliness. To make sure that the spaces I create promote well functioned for those who live in there . My aim is always to ensure movement within the space to maximizing functionality and bringing in plenty of natural light and fresh air. Considering the climate's key , I incorporate design techniques like positioning, shading elements and natural airflow to work in harmony with the local weather conditions aiming for both comfort and sustainability. My goal through design is to establish living spaces that cater to residents needs while also nurturing a feeling of health and connection, with their environment.

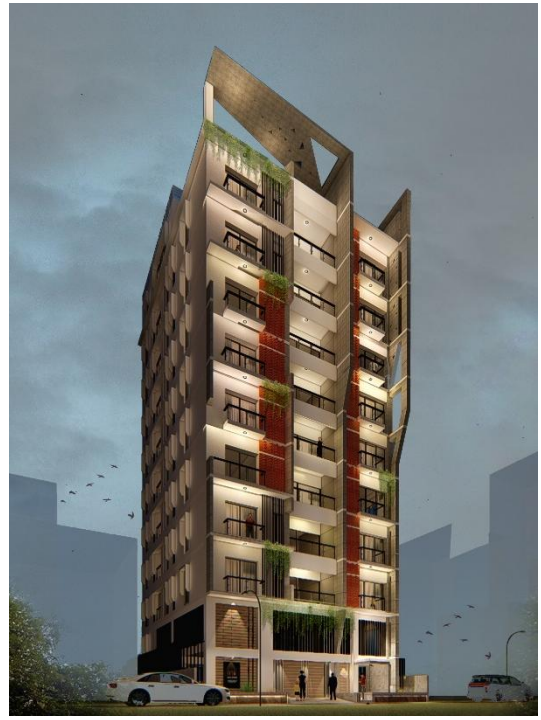


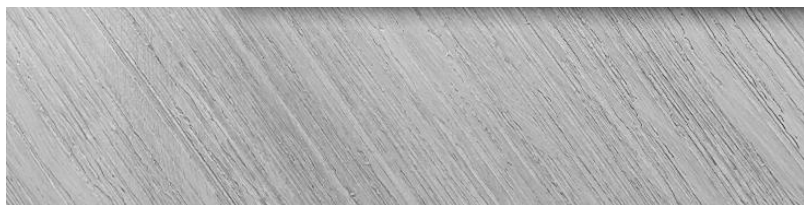
PLAN | RENDERS



1ST FLOOR(SINGLE UNIT)

SCALE:1:100





I N T E R I O R W O R K

Professional Work , Year | 2021

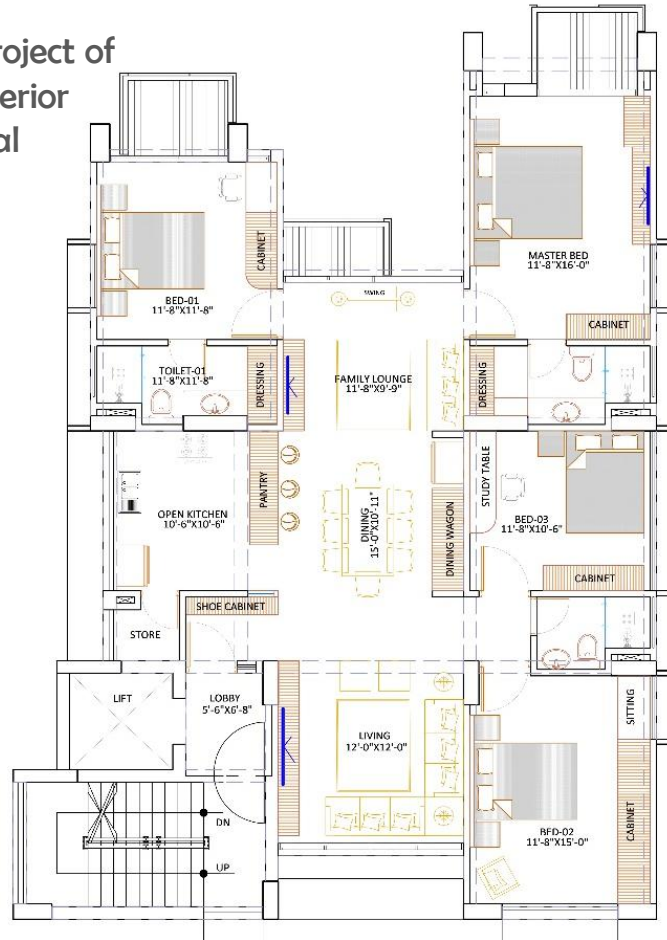
Location: Dhaka , Bangladesh

Designation : Chief Architect

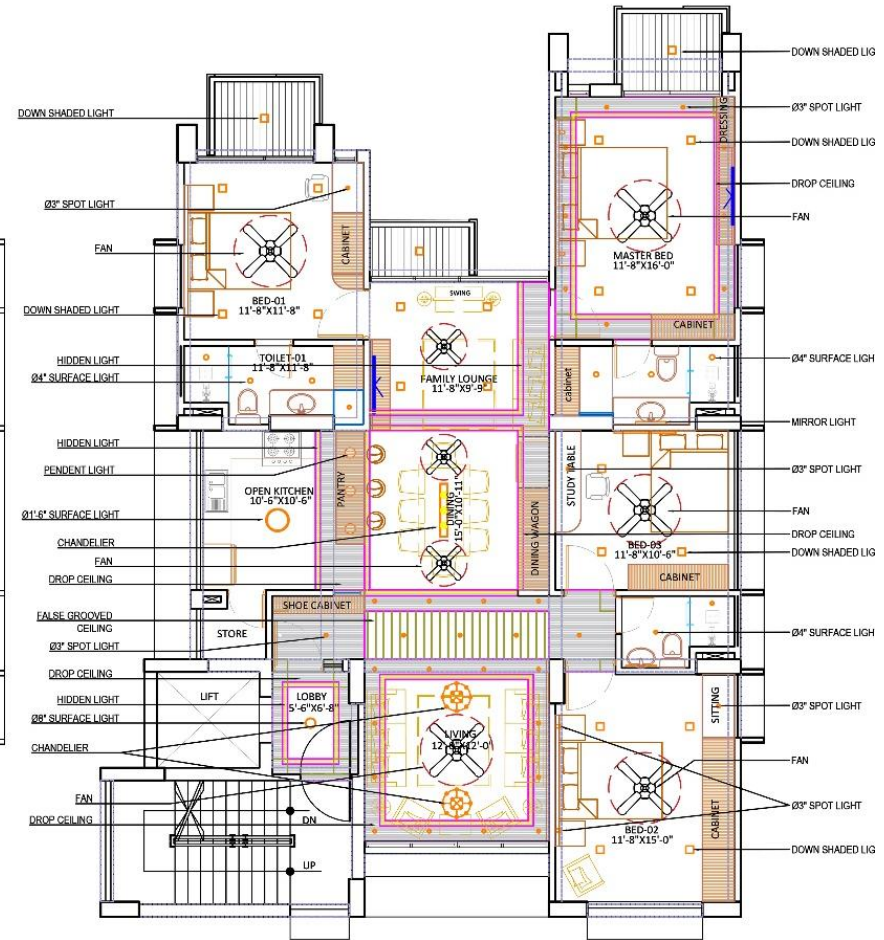
Project type : Complete Project of
Residence interior
& Commercial

OVERVIEW

In terms of interior design , I am into minimalism . The main key points of mine are bare essentials, open layout, clean lines, minimalist furniture, clutter-free surfaces, neutral colors , and simple textures. The another major concept of mine is to add green to a home to bring the outside, in! Placing a few houseplants throughout home can actually reduce carbon dioxide levels and cut down on airborne dust.



FURNITURE LAYOUT PLAN(OPTION-01)



CEILING LAYOUT PLAN(OPTION-01)

INTERIOR WORK

RESIDENCE INTERIOR

Handover Pictures



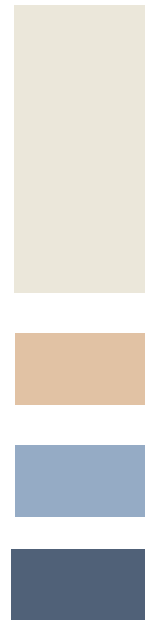
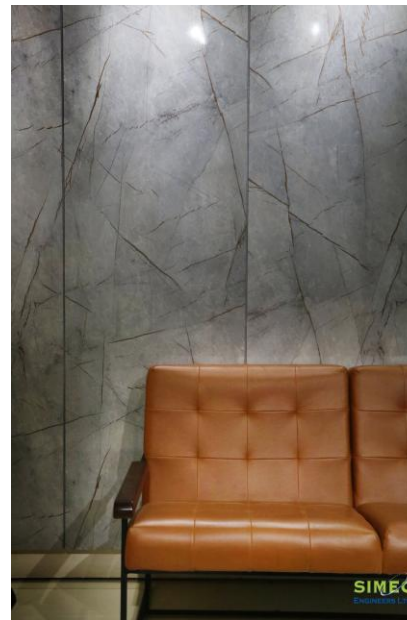
COMMERCIAL OFFICE INTERIOR

Handover pictures



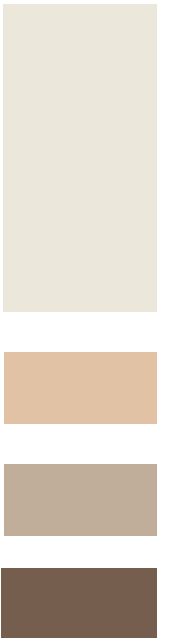
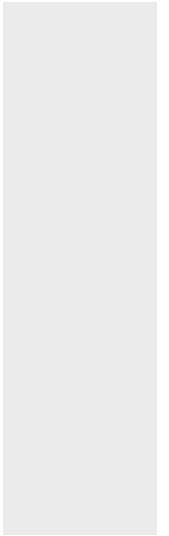
COMMERCIAL OFFICE INTERIOR

Handover pictures



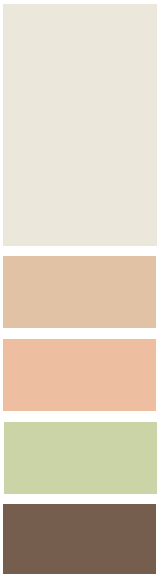
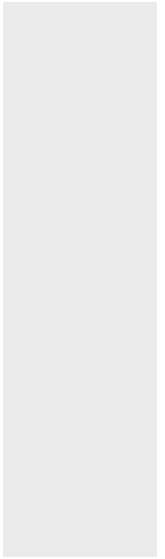
HOTEL GUEST FLOOR INTERIOR

Handover pictures



HOTEL GUEST FLOOR INTERIOR

Handover pictures



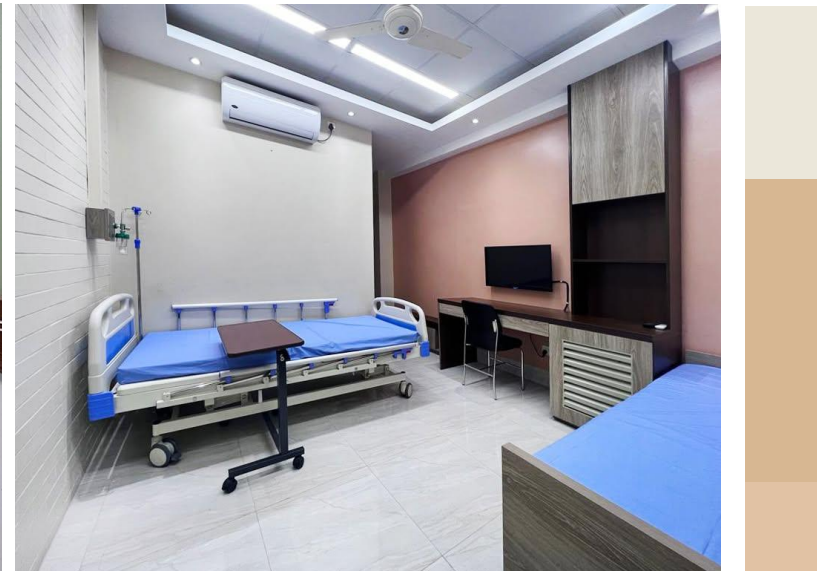
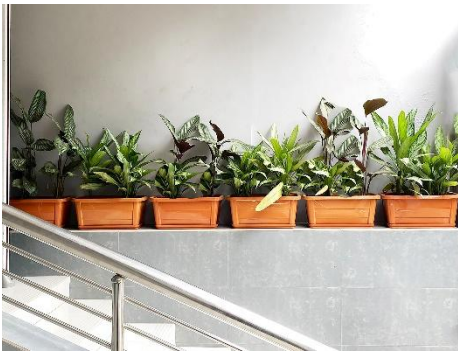
COMMERCIAL OFFICE INTERIOR

Handover Pictures



HOSPITAL INTERIOR

Handover Pictures





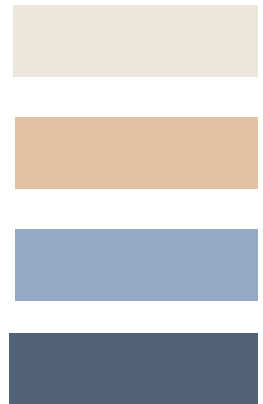
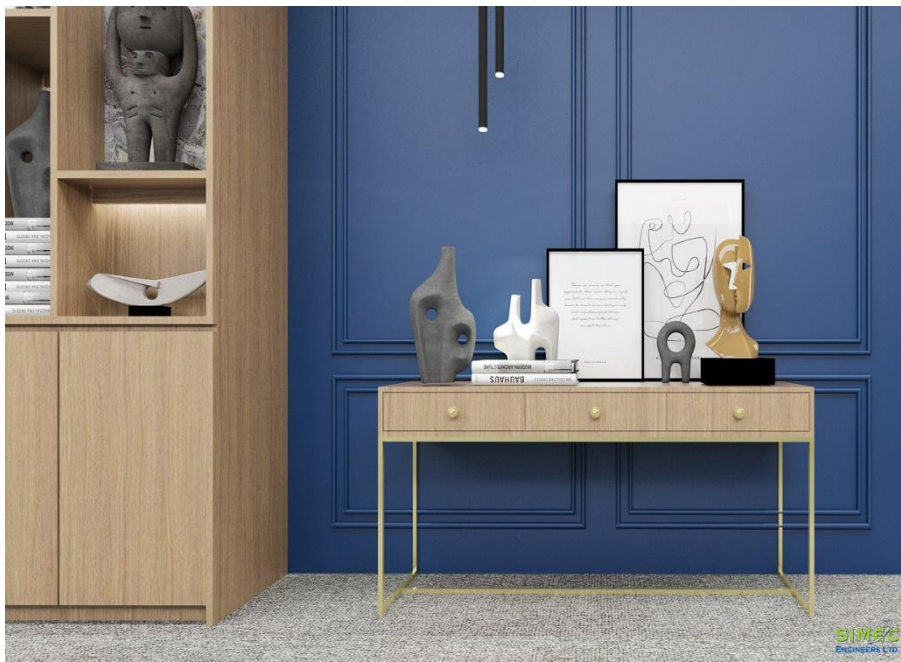
RESIDENCE INTERIOR

Render Image



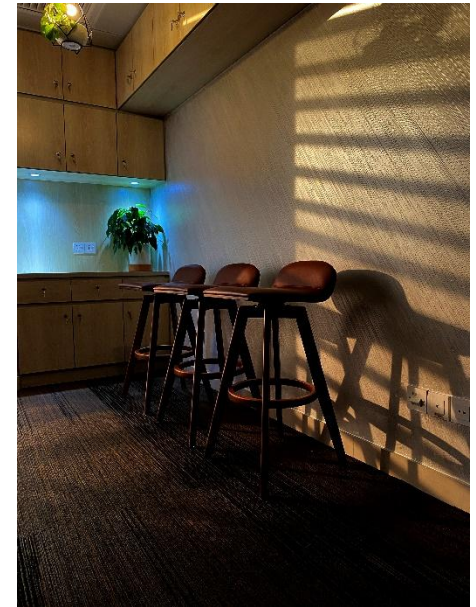
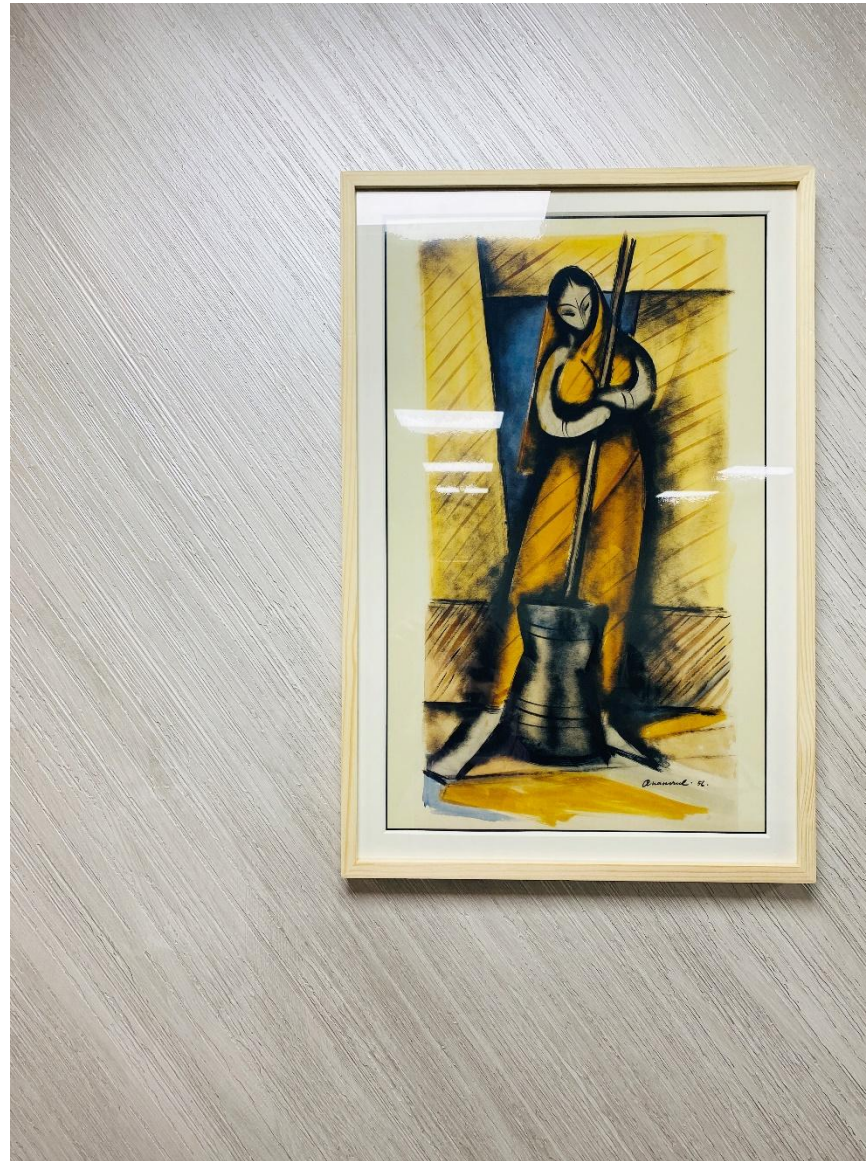
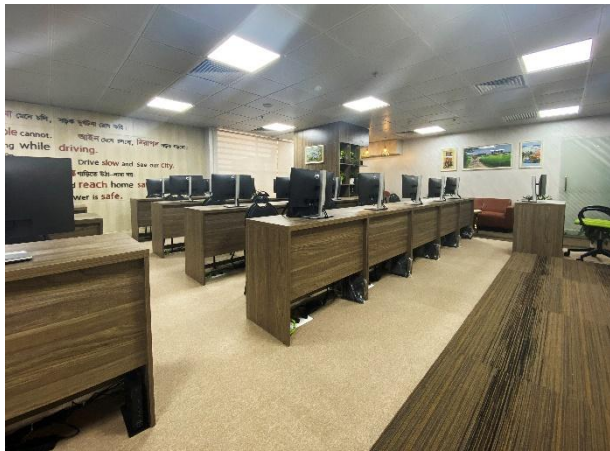
COMMERCIAL OFFICE INTERIOR

Render Image



BRTA LAB INTERIOR

Handover Pictures





L A N D S C A P E W O R K

LANDSCAPE

Professional Work ,
Year | 2021

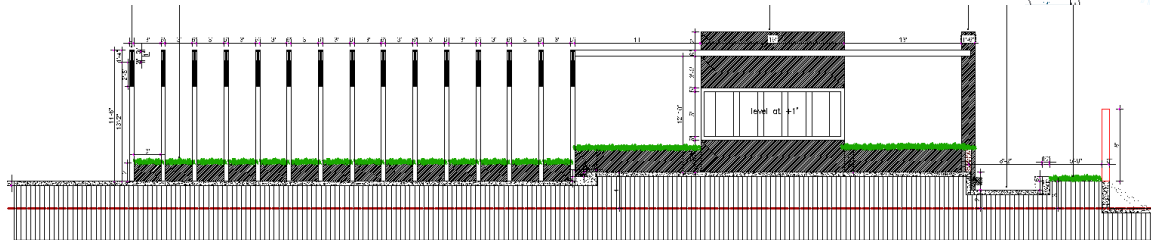
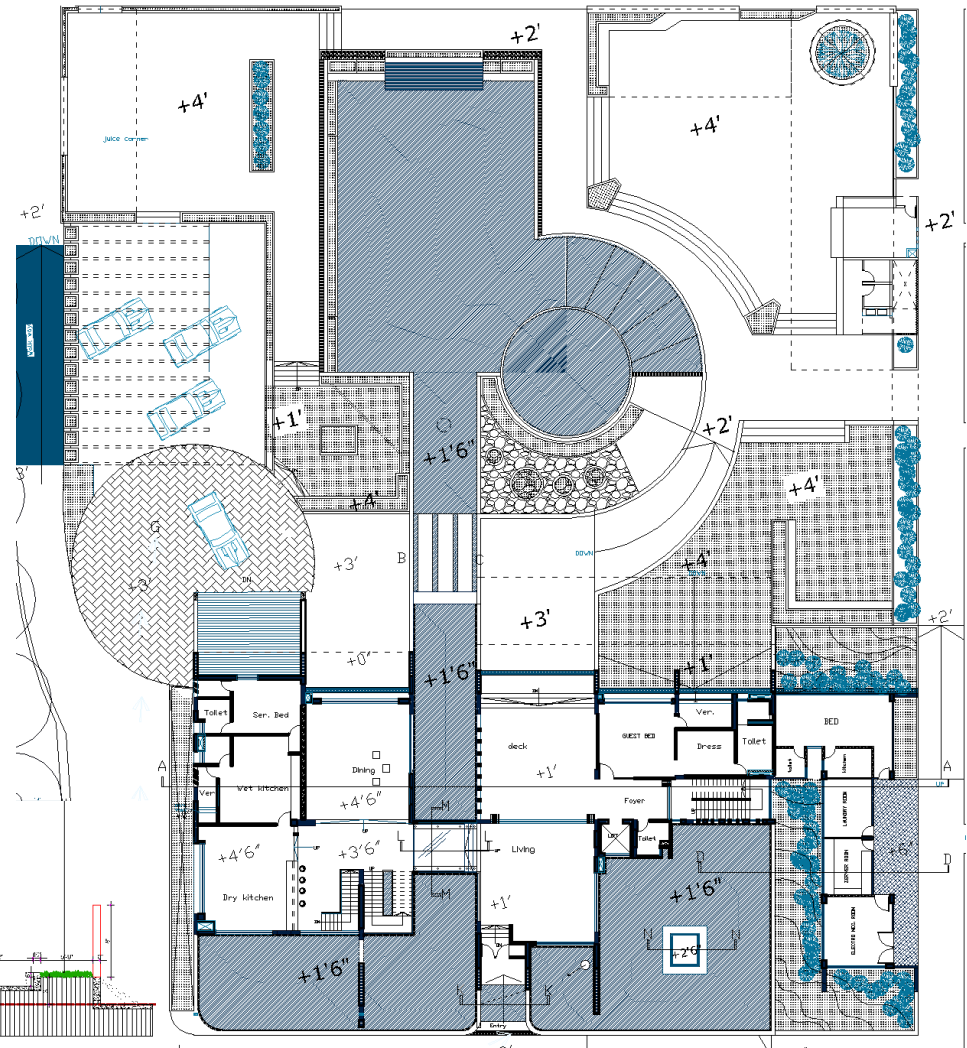
Location: Dhaka , Bangladesh

Designation : Chief Architect

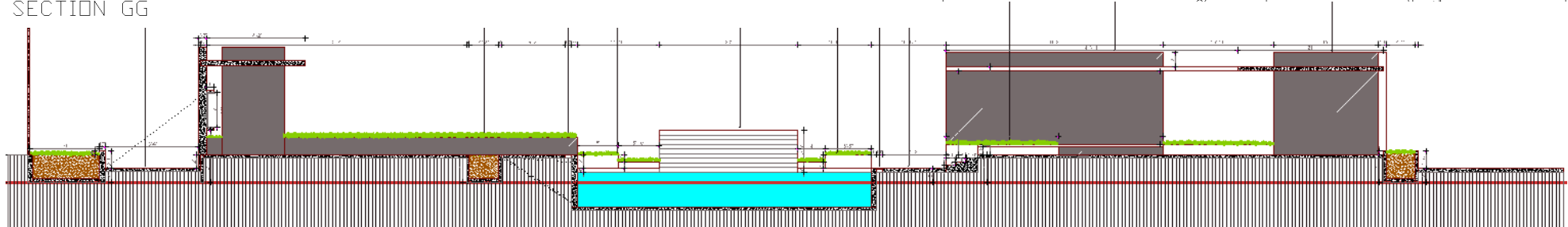
Project type : Landscape

OVERVIEW

In my design my aim was to create a base plan which expresses how the landscape will work in terms of function and flow. It tells the story of how the clients can use the new spaces and the wonderful features which have been designed in. Contrast is an another important element in my design. It can make all the difference between a boring landscape or a landscape that will pop. Contrasting textures, contrasting colors, contrasting plant sizes , contrasting leaf structures , level variation and blending green and water all add interest.



SECTION GG

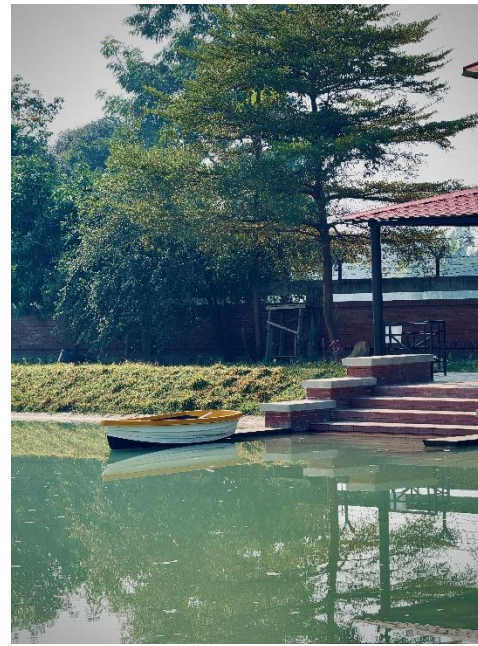


SECTION HH

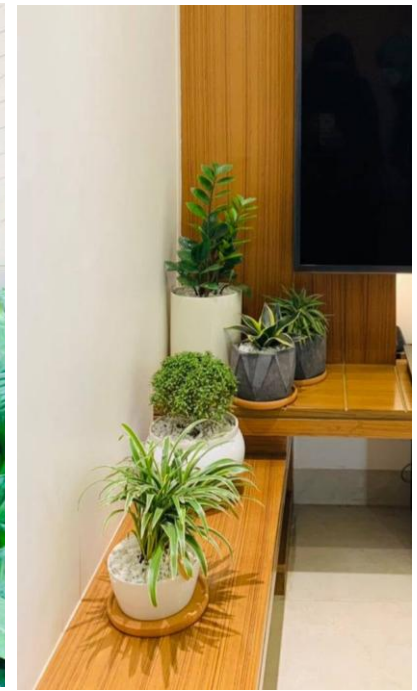
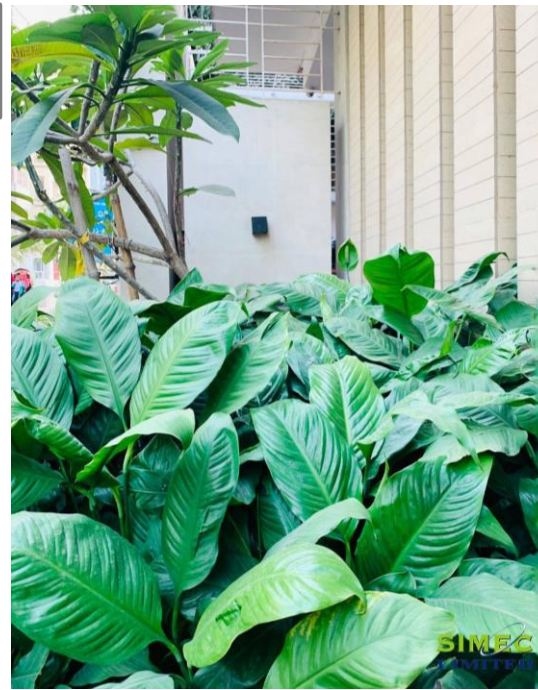
RENDERS

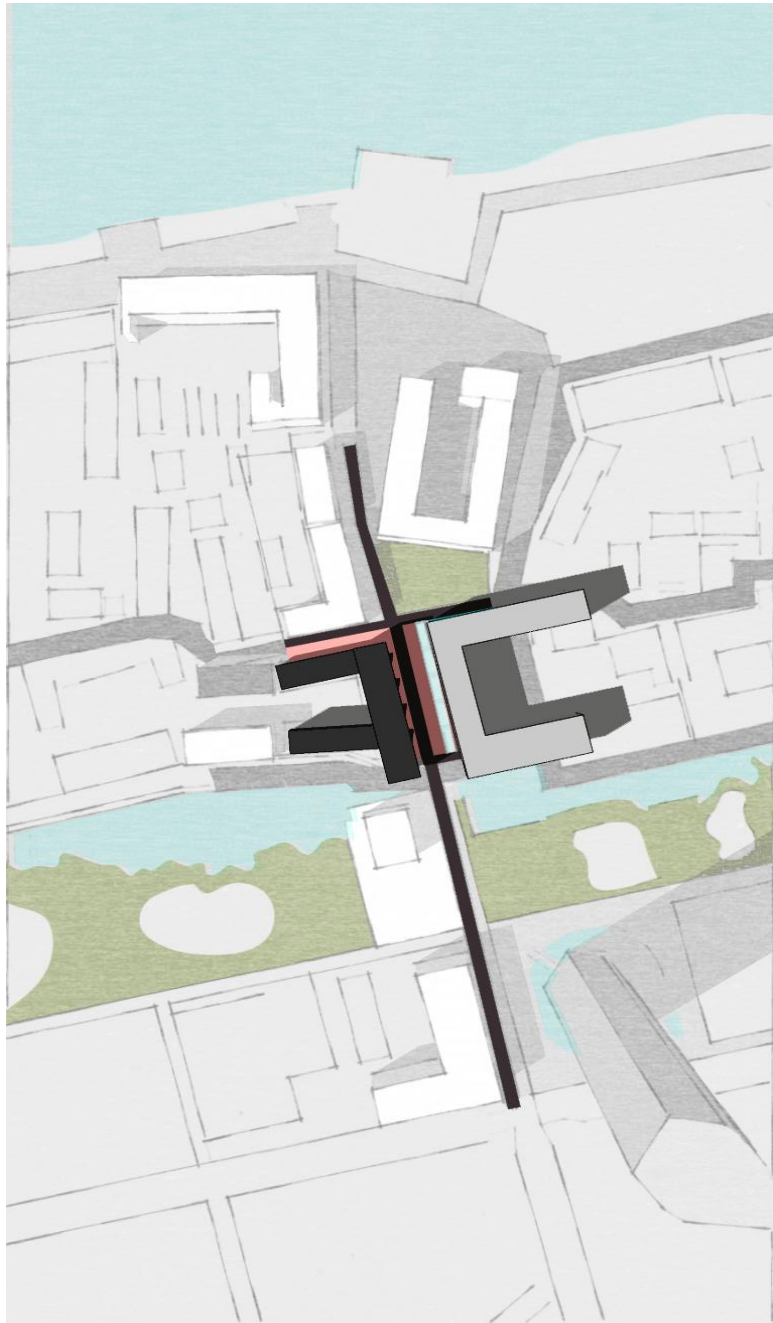


HANDOVER PICTURES



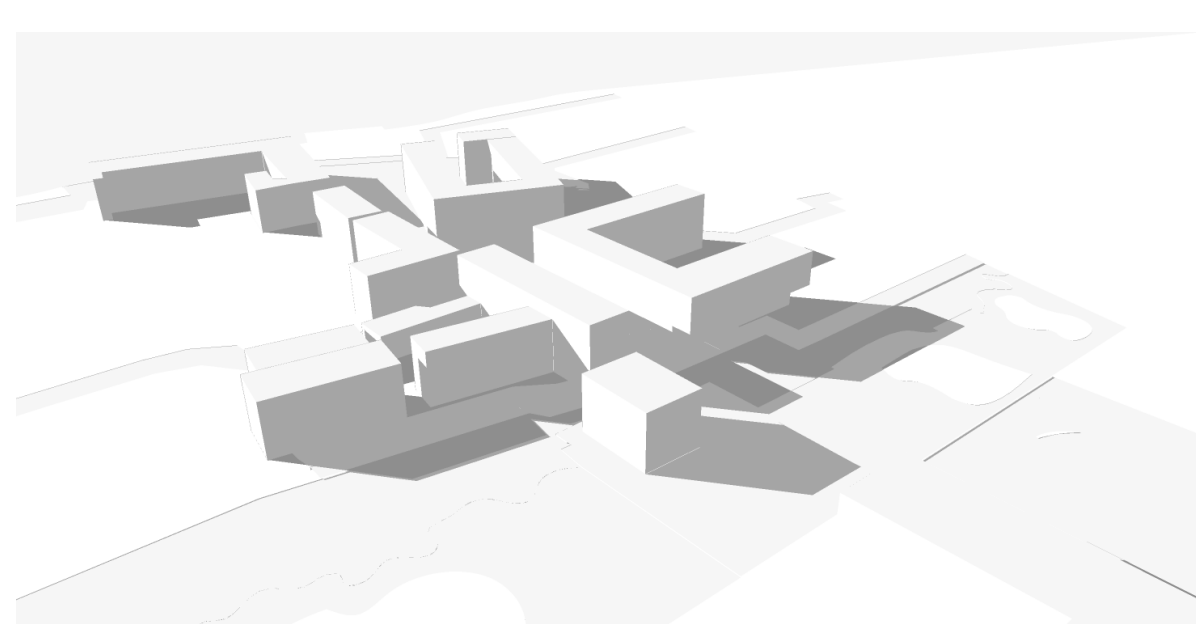
HANDOVER PICTURES





A C A D E M I C P R O J E C T S

MASTER'S PROGRAMME
(ENERGY EFFICIENCY AND ENVIRONMENTAL
BUILDING DESIGN)
LUND UNIVERSITY



Impact of facade colour on visual preference, visual comfort, and outdoor thermal comfort in a Swedish urban environment

Degree project, 2026

Contributors: Issana Tasnim, Tahseen Farhana

PROJECT OVERVIEW & PROBLEM

As cities become denser, natural landscapes are increasingly replaced by hard, heat-absorbing surfaces, affecting daylight, urban temperatures, and pedestrian comfort. This study explores how facade colour—hue, saturation, value, and reflectance—affects both visual preference and urban microclimate in Sweden.

Using a mixed-method approach, the research combined 182 participant surveys, on-site reflectance measurements, and CitySim Pro simulations across six case-study buildings to examine visual preference, perceived comfort, surface temperature, and outdoor thermal comfort.

APPROACH

Developed a 10-colour Nordic facade palette and tested black (7% reflectance), grey (23%), and white (81%) scenarios under varied orientations and seasonal conditions. Compared demographic preference groups and evaluated MRT/PET, surface temperature, and annual energy performance.

DESIGN TAKEAWAY

Facade colour is not an isolated aesthetic choice — orientation, solar exposure, season, geometry, user perception, and energy performance must all be considered together.

KEY RESULTS SUMMARY

28.7°C

surface temperature difference between black (67.2°C) and white (38.5°C) facades under solar exposure.

166–184 kWh/m²

Annual internal heating demand range across facade colours and reflectance scenarios

182 Participants

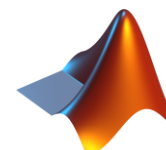
Survey respondents across demographic groups — preference varied by cultural background



CITYSIM PRO



SKETCHUP



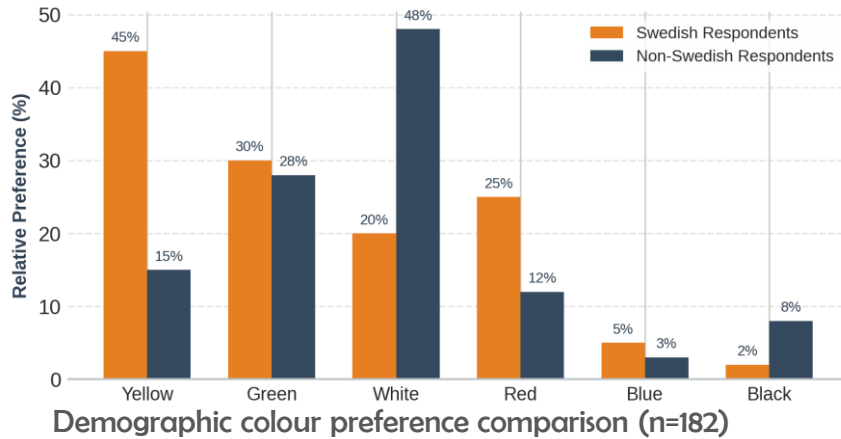
MATLAB



LADYBUG

Visual Evidence — Research Results

Visual Preference across Facade Color Schemes



Demographic Colour Preference

Preferences were clearly differentiated by cultural background, showing that facade colour is interpreted through both place and identity.

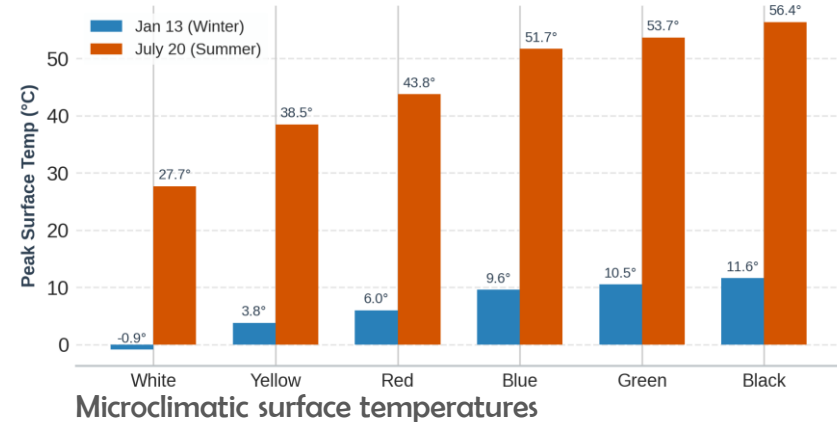
Swedish participants favoured yellow more strongly, while non-Swedish participants leaned toward white.

The result reinforces that visual comfort and aesthetic acceptance are not universal — they vary across user groups and should inform design decisions early.

- **Visual Comfort vs. Preference:** Although people preferred yellow, White scored highest for visual comfort as it is perceived as 'soft' and 'soothing', reducing eye strain.

Outdoor thermal comfort & thermodynamics

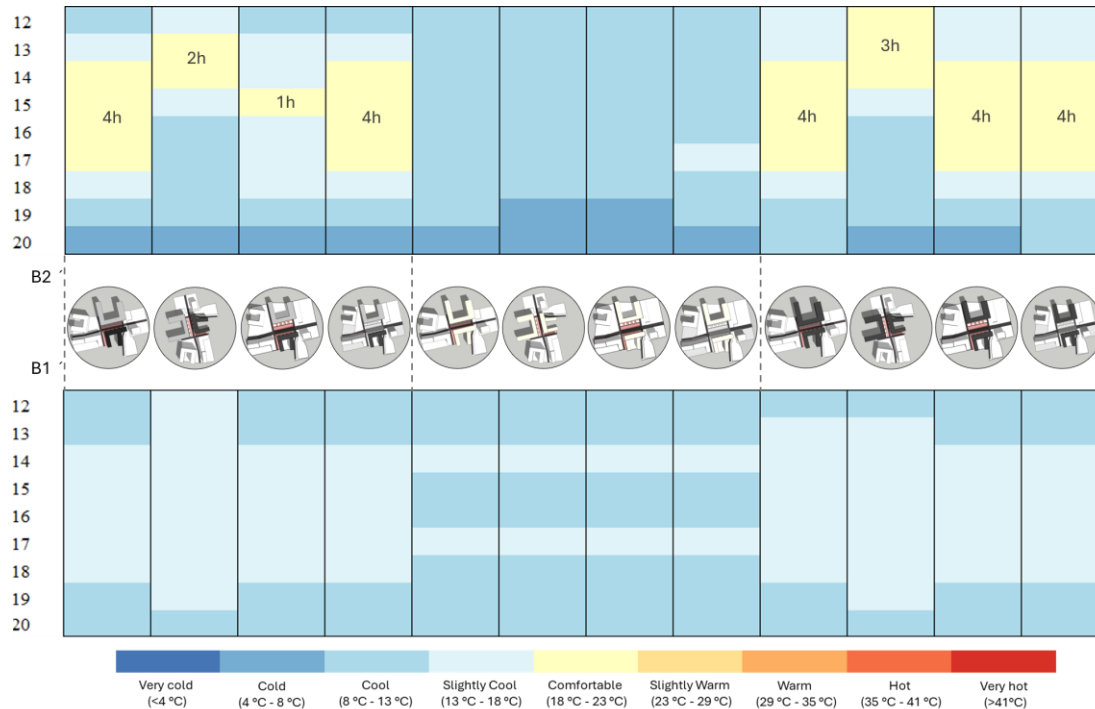
Microclimatic Impact: Solar-Exposed Facade Peak Temperatures



Direct Solar Radiation Drives Huge Thermal Differences

- During Peak Summer (July 20), there is a massive 28.7°C difference in surface temperature between the lowest reflectance (black: 56.4°C) and highest reflectance (white: 27.7°C) facades.
- During Winter (Jan 13), solar elevation is low, making the temperature variance much tighter (black: 11.6°C vs. white: -0.9°C), illustrating that facade color reflectance has negligible thermal comfort impact in winter.

Outdoor thermal comfort & thermodynamics



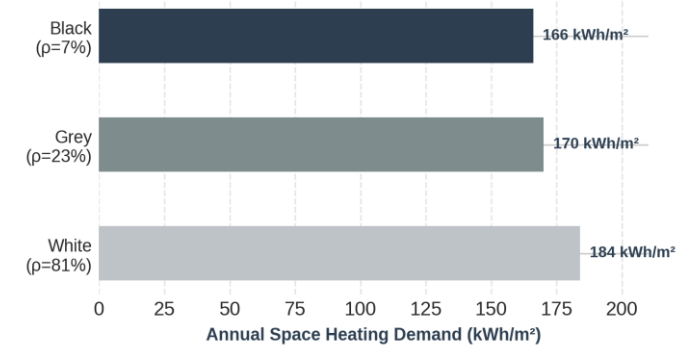
Outdoor microclimatic performance was simulated in CitySim Pro using Copenhagen-Kastrup weather data (55.6°N). PET was calculated for walking (270 W) and seated (108 W) pedestrians under real wind conditions.

PEDESTRIAN SUMMER THERMAL COMFORT

In high-latitude Swedish climates, strong winds and cool ambient air create localised cool stress for seated pedestrians, even in July. High-reflectance white facades remain cool (27.7°C) but fail to warm the microclimate. Conversely, low-reflectance black facades absorb solar energy, reaching surface temperatures of 56.4°C. This heat is re-emitted as longwave radiation, increasing the MRT and raising localized PET to the comfortable 18°C–23°C range for 38% of the afternoon (3.0 comfortable hours).

Annual heating demand

The Nordic Energy Paradox:
Low Reflectance (Black) Facades Save 10% Heating Energy



The Swedish energy paradox

Low Reflectance Facades Act as Winter Thermal Drivers:

- In cold Nordic climates, low reflectance facades (black) absorb shortwave solar radiation and passive solar heat, which transfers into the building envelope.
- This thermal transmission reduces the space heating load of buildings by 10% annually compared to reflective white facades (166 kWh/m² for black vs. 184 kWh/m² for white).

Executive Policy & Design Guidance

Facade colour should be treated as a microclimatic and thermodynamic design parameter, not only an aesthetic choice.

- **Balance Performance:** White/yellow improve visual comfort but increase heating demand by ~10%, while black improves winter heating but can reach 56.4°C in summer.
- **Respond to Orientation:** Use darker colours on south façades for winter solar gains and lighter colours on north façades to enhance daylight with minimal heat gain.
- **Enhance Pedestrian Experience:** Avoid monotonous colour schemes and integrate varied façades, greenery, and water to improve visual quality and summer comfort.



Västra Karup Församlingshem — Adaptive Reuse & Solar Retrofit

Parametric simulation | Lca + Lcc | Photovoltaics | Adaptive reuse | Pareto optimization

AEBN21 - FALL 2025

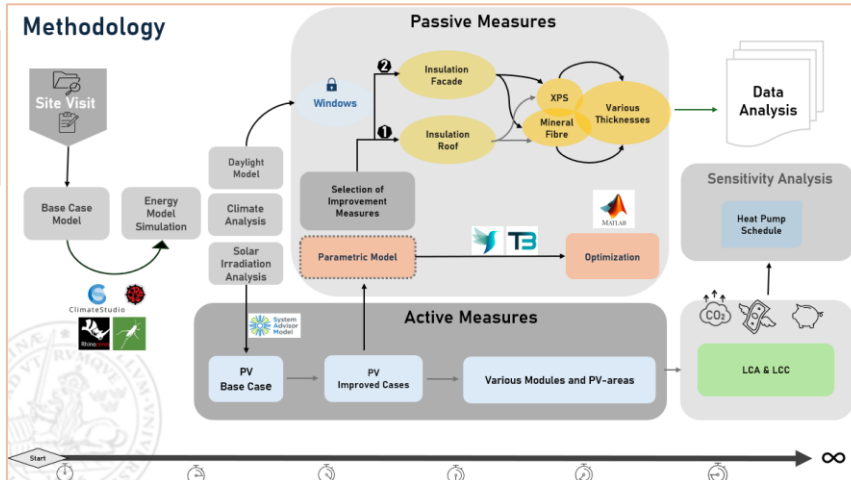
Contributors: Issana Tasnim, Tahseen Farhana, Anna Maria, Savo Soleman

PROJECT OVERVIEW & PROBLEM

Europe's ageing public building stock faces urgent requirements to improve energy performance and life-cycle efficiency. This portfolio examines the retrofitting of the Västra Karup församlingshem, a public building in southern Sweden with 474 m² of heated area across two storeys, including an attic and basement. The existing building (Base Case) exhibits substantial thermal losses due to highly uninsulated envelopes, resulting in an annual energy demand of 29,838 kWh based on the utility data. The envelope consists of an 800 mm stone facade for the main hall with a U-value of 2.69 W/(m²·K), a dining hall facade with a U-value of 1.76 W/(m²·K), and an attic with a U-value of 0.95 W/(m²·K). Daylight conditions are also inadequate, with mean Daylight Factor values of 0.7 % on the ground floor and 0.3 % on the first floor, both below the Miljöbyggnad 4.0 bronze requirement of 0.8 %.

APPROACH

The base case was calibrated in Rhino 8 against site-survey data, then evaluated through Climate Studio and Ladybug-Honeybee for energy and daylight, and SAM for PV yield. Passive measures (XPS and mineral-fibre insulation at 200/300/400 mm, new triple glazing) and active measures (rooftop PV sized against summer demand) were combined parametrically — 432 simulated combinations in total — across two functional scenarios shortlisted from three adaptive reuse proposals: office and recreational centre. Every combination was scored for energy demand, 50-year life cycle cost and cradle-to-grave life cycle GWP, then filtered to a Pareto-optimal set in MATLAB.

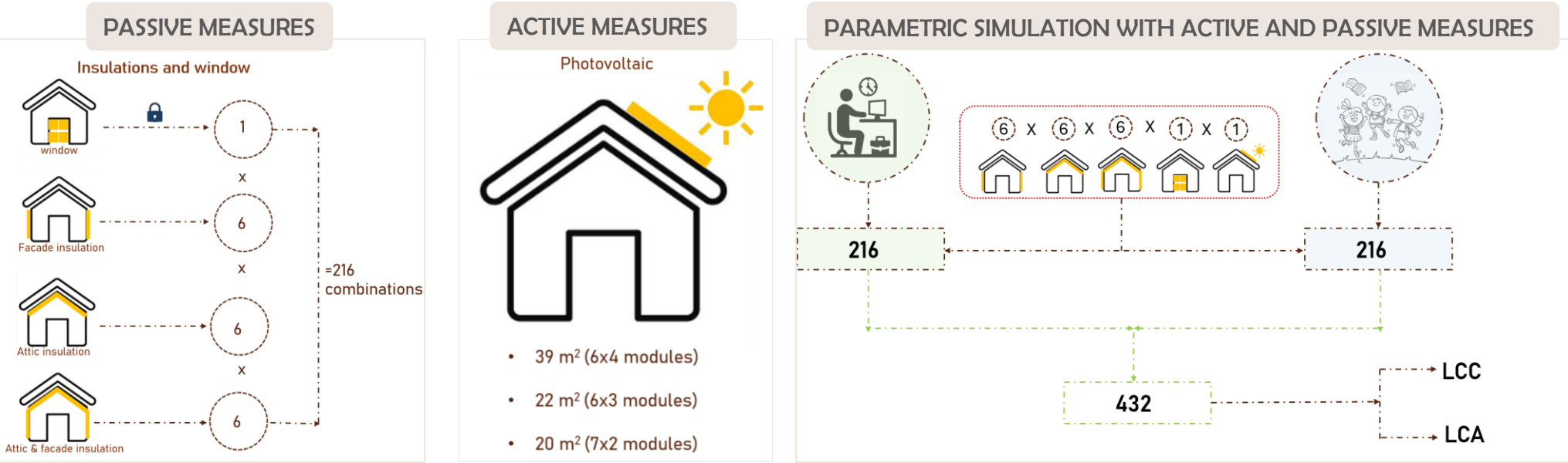


DESIGN TAKEAWAY

Combining passive measures (U-values optimized to 0.065-0.105 W/(m²·K)) with local solar energy (7 kWp PV) yields a massive performance upgrade. Functional analysis confirms that while both options operate at exceptional efficiency, the Recreational Centre offers superior lifecycle economic and carbon benefits over the Office function due to lower occupancy densities and tailored heating schedules.

PARAMETRIC STUDY

Parametric simulations of passive and active measures were conducted for the office and recreational centre. Six insulation variations were applied across three building-zone components, producing 432 combinations per function. All scenarios were run using Colibri and the TT toolbox, with outputs sorted in Excel and processed using basic programming. The simulations identified optimal insulation thickness and type, along with suitable PV sizes, while glazing parameters remained fixed based on daylight requirements.



KEY RESULTS SUMMARY



**ENERGY DEMAND
(WITHOUT PV)**
58.0 → 31.6 → 26.3
Kwh/m2/year, Base case VS.
Office vs.
Recreational centre reuse



**ENERGY DEMAND
(WITH PV)**
37.0 → 19.7 → 16.8
Kwh/m2/year, Base case VS.
Office vs.
Recreational centre reuse

7 → 6 → 4
kWpeak, Base case VS.
Office vs.
Recreational centre reuse

0.7% → 3.5%
Mean DF achieved, Ground floor
0.3% → 2.6%
Mean DF achieved, 1st floor

PHOTOVOLTAIC INTEGRATION — OFFICE

With the building reprogrammed as an office, the south-east-facing roof receives up to 859 kWh/m² of annual solar irradiation. PV sizing was optimised in SAM for summer conditions to prevent overproduction, and without battery storage, all surplus is exported to the SE4 grid while deficits are purchased in real time. The Office layout uses eighteen Sunpower SPR-X21-355 panels (20.55 % efficiency, 335.21 Wdc) in a 6×3 array covering 29.56 m² with a 6 kW_p capacity, paired with a SolarEdge SE6000x inverter (96.85 % efficiency, 5330 Wac) requiring one replacement over the 50 years.

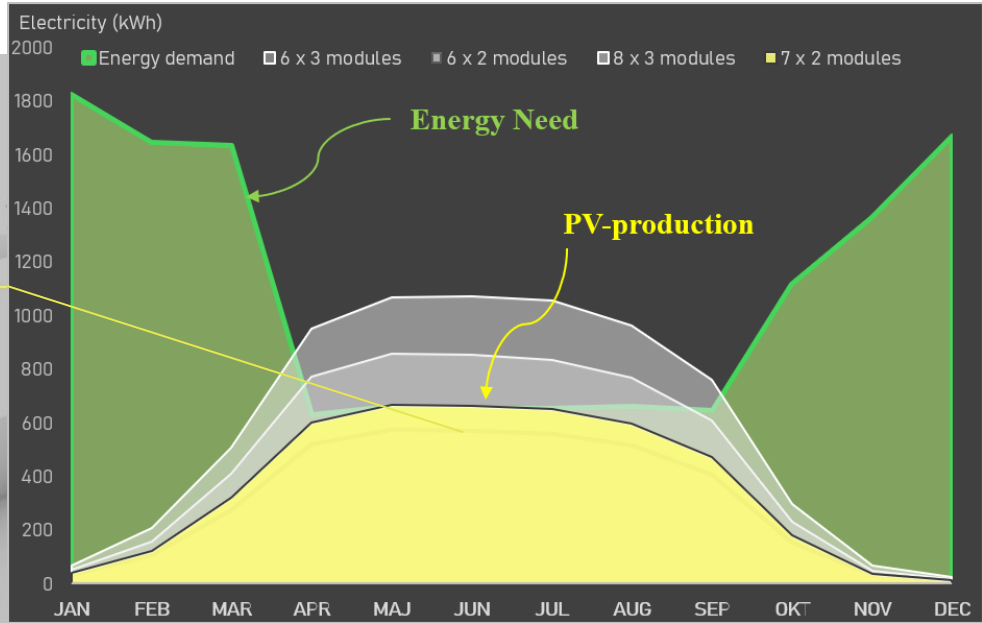
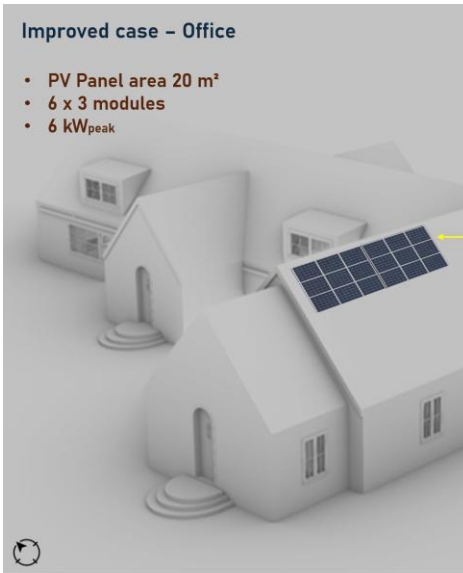
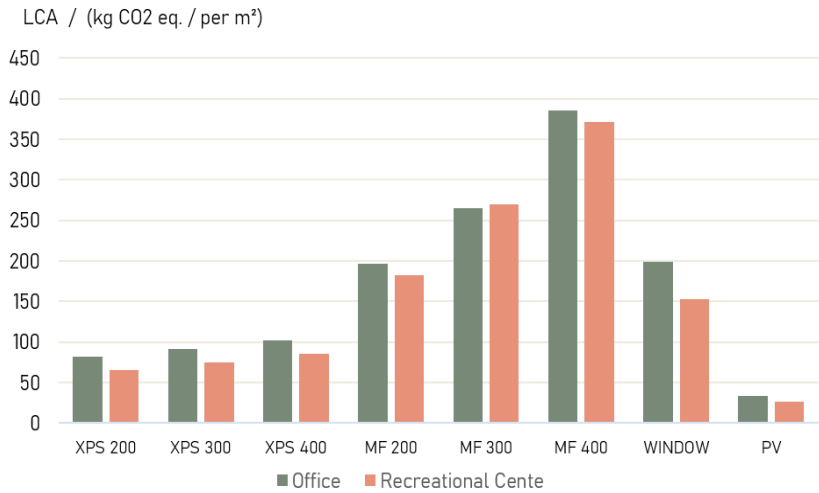


FIG. — Office energy demand vs. modelled PV production across module configurations. The chosen 6×3 array covers the bulk of spring–summer load, leaving winter demand to the grid.

LIFE CYCLE COST (LCC) AND LIFE CYCLE ASSESSMENT (LCA)

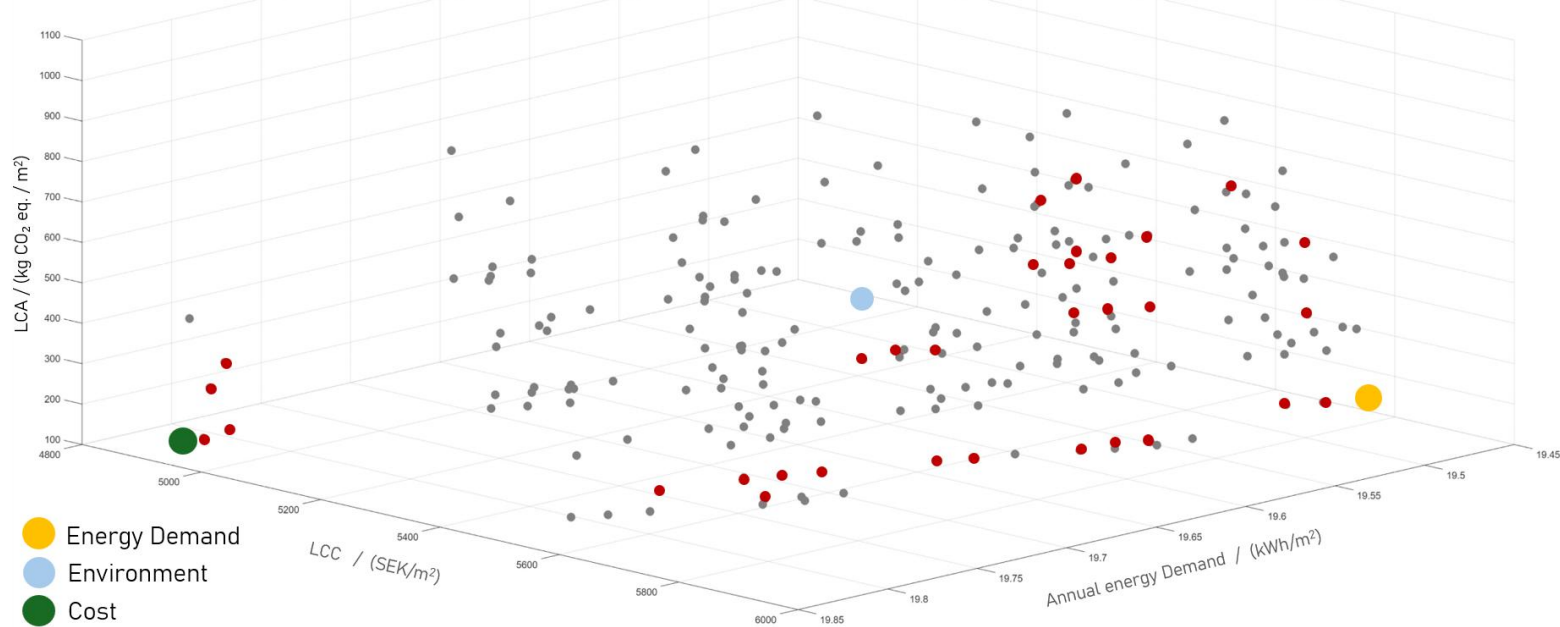


LCC analysis assesses total economic performance over 50 years. Initial costs for windows, insulation materials, and labour were sourced from Swedish Wikells databases, while PV system costs were calculated per module. Net operational costs were calculated by multiplying hourly electricity deficits and surpluses with Swedish spot prices and discounted at 2.0 %. LCA quantifies GWP in kg CO₂ eq. using Scandinavian EPDs compliant with EN 15804+A2 across all stages (A1–C5). Under combined passive/active evaluation, PV yields the lowest LCC and environmental impact.

FIG. — Left: 50-year life cycle cost by measure (M SEK). Right: cradle-to-grave Global Warming Potential by measure (kg CO₂-eq./m²).

MULTI-CRITERIA DECISION ANALYSIS & PARETO FRONT

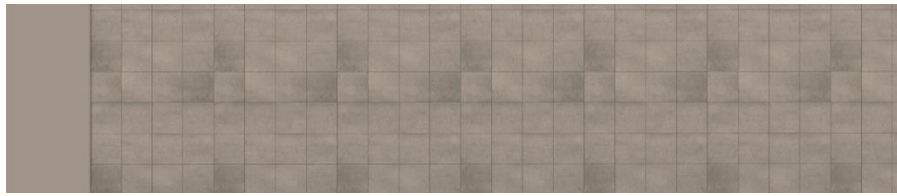
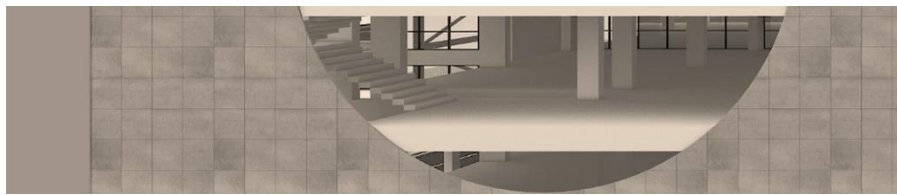
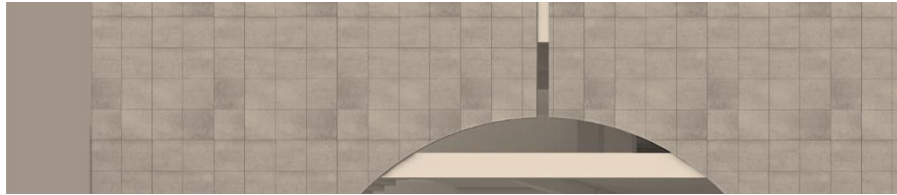
Multi-criteria optimisation using a Pareto Front was formulated to isolate combinations that are 'better in every possible way'. Out of 432 parametric simulations for the Office function, 34 Pareto optimal solutions were identified (compared to 37 for the Recreational Centre). These represent the mathematical boundaries where energy demand, LCC, and LCA are co-optimised. Lowering energy demand requires added insulation thickness, which increases initial LCC and embodied LCA (GWP), but reduces operational costs. The Pareto optimisation isolates the precise configurations of envelope layers that resolve these competing parameters.



MAIN HALL — ROOF	OFFICE GROUND FLOOR — FAÇADE	OFFICE FIRST FLOOR — FAÇADE + ROOF	BEST FOR ENERGY	BEST FOR COST	BEST FOR ENVIRONMENT
400 mm XPS	300 mm Mineral Fibre	300 mm Mineral Fibre	✓		
400 mm XPS	200 mm XPS	200 mm XPS		✓	
200 mm XPS	200 mm XPS	200 mm XPS			✓

Retrofit Design Guidelines for Historical Public Buildings in Sweden

- **Balanced decision-making** — Retrofits must consider EUI, LCC, GWP, and thermal comfort through multi-criteria optimisation.
- **Local solar synergy** — Size PV to summer demand; in SE4, a 6 kWp monocrystalline system gives optimal export/import without batteries.
- **Insulation limit ~200 mm** — First 200 mm cuts demand by 2.4%; the next 300 mm adds only 0.7% (clear diminishing returns).
- **Dynamic ventilation + HR** — Use mechanical ventilation with ≥80% heat recovery, reducing infiltration to 0.10–0.15 ACH.



A C A D E M I C P R O J E C T S

BACHELOR OF ARCHITECTURE

MARITIME MUSEUM

THESIS PROJECT

Academic Project : Thesis project | 2014

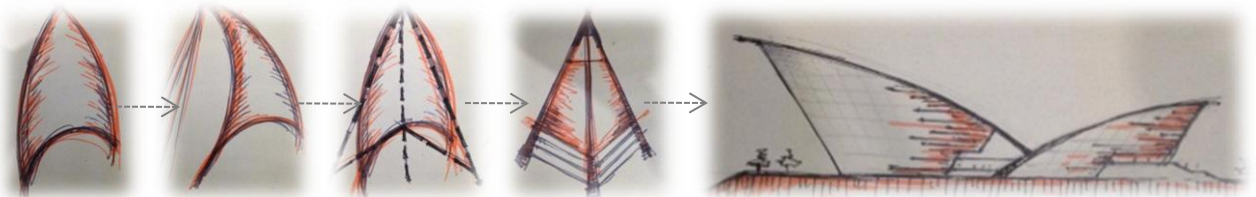
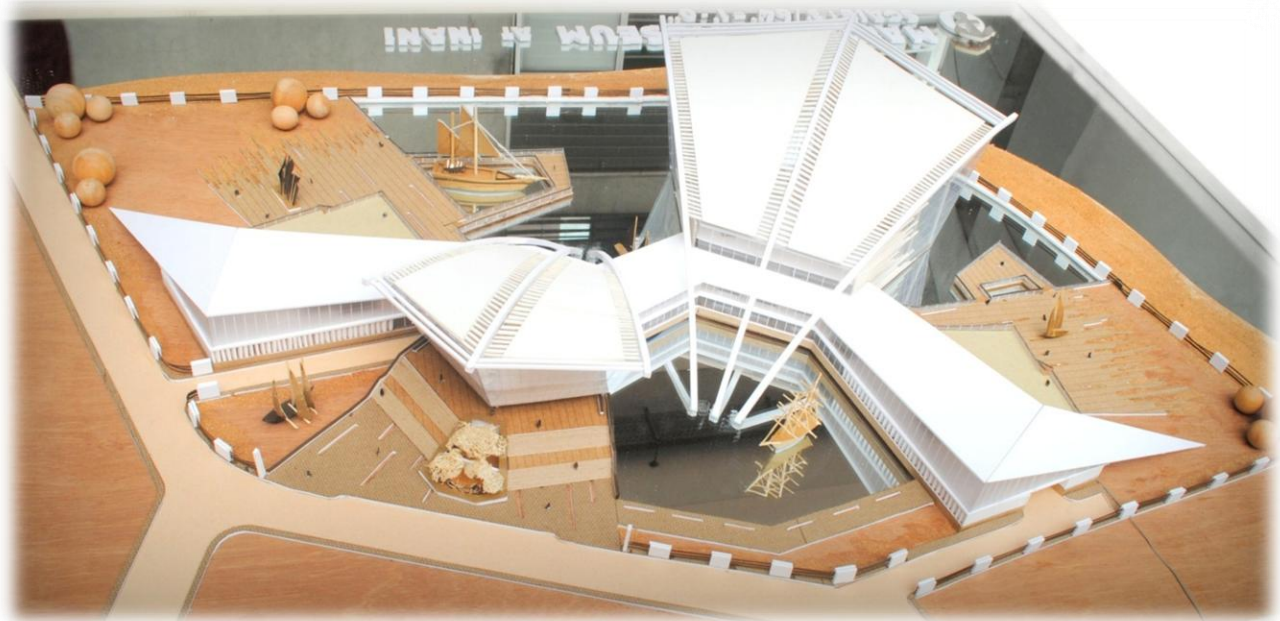
Location: Chittagong , Bangladesh

Instructor : Dr M. A. Muktadir

Project type : Recreational

OVERVIEW

The requirement for the site was to accommodate a museum that not only displays an extensive maritime collection, but also connects the town with the sea and creates a destination. In my design the museum has its feet in the water. Built on pile foundations. My concept derives from the “SAIL ON BOAT”. It is in essence comprised of two parts: under the big sail I accommodated all the exhibits and under the small sail there were all the administrative function and conference hall. My aim was to make this museum a landmark between the city and the sea life and that's why I made the boundary blur to blend the structure with the sea.



CONCEPTUAL SKETCH



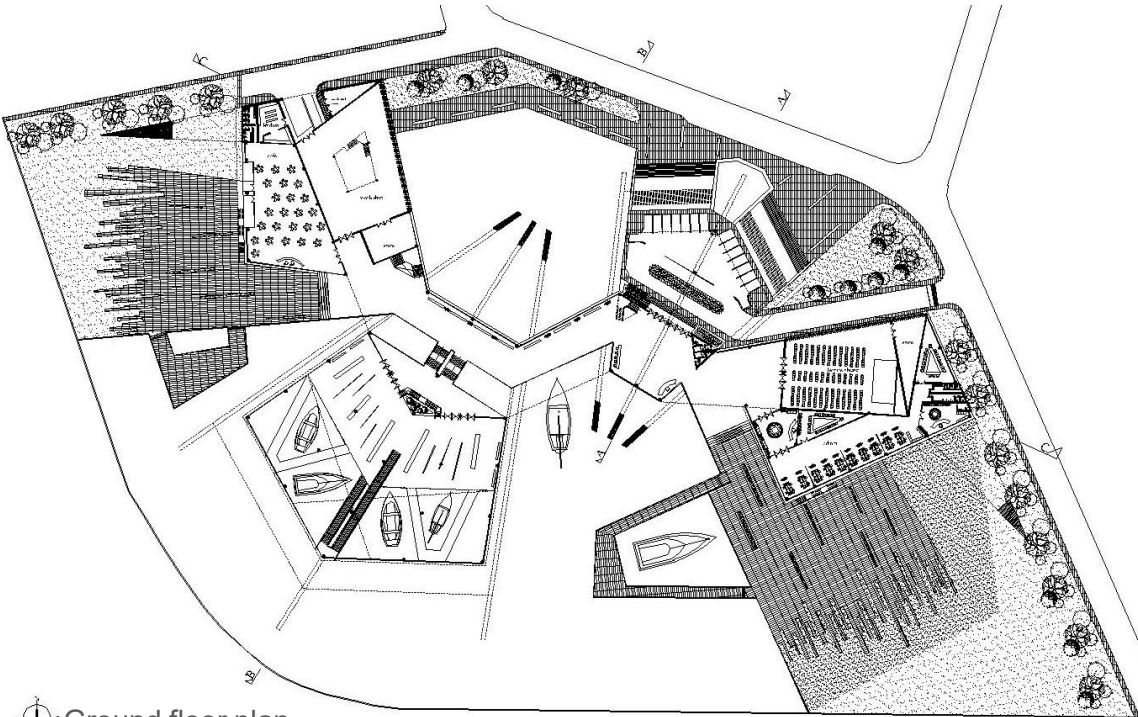
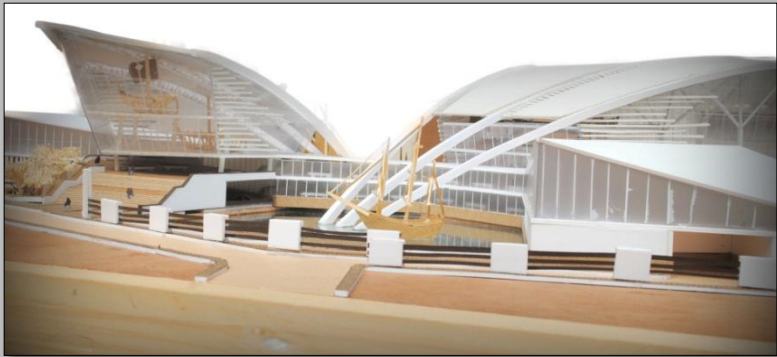
PLAN | ELEVATION | VIEWS



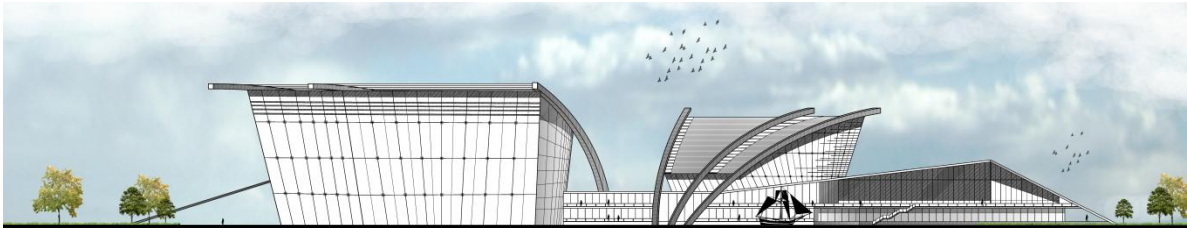
VIEW FROM ENTRY



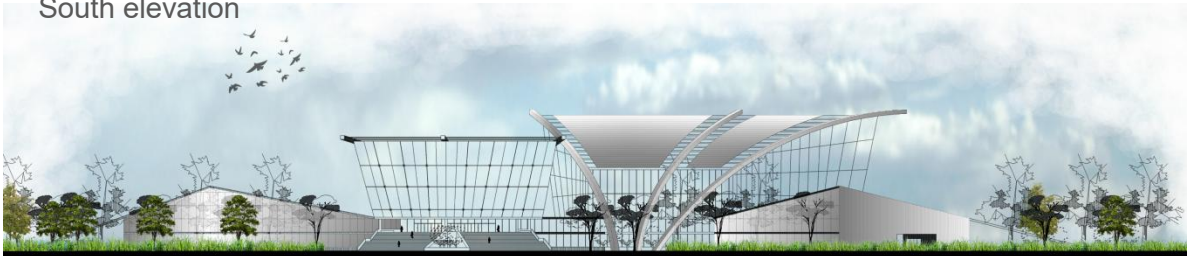
VEHICULAR ENTRY



Ground floor plan

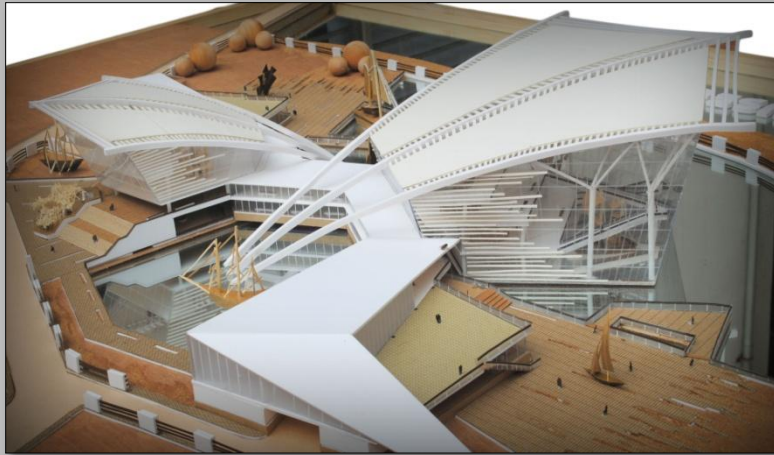


South elevation



North elevation

PLAN | ELEVATION | VIEWS



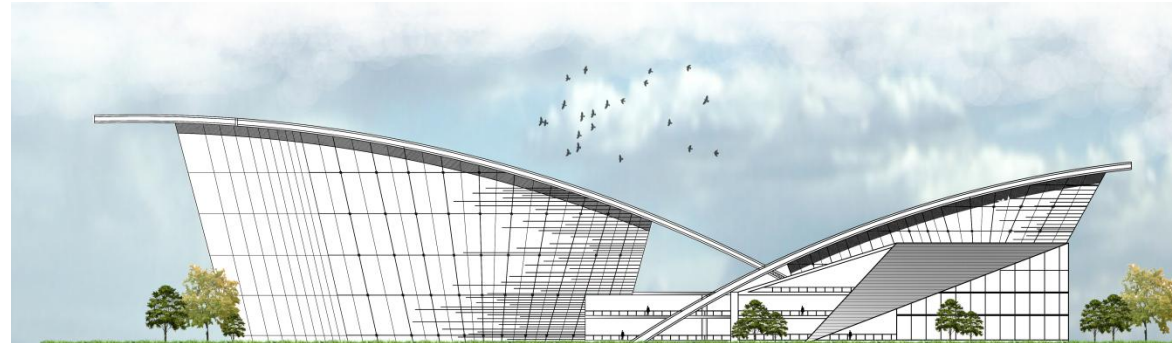
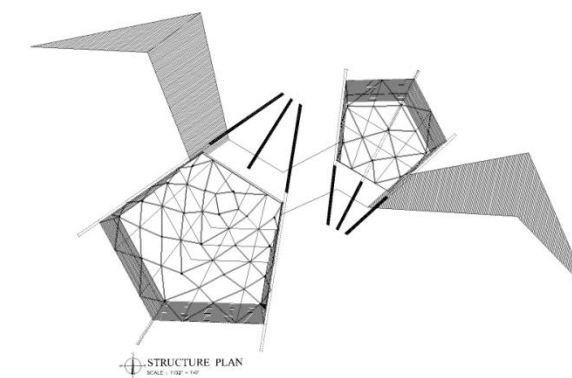
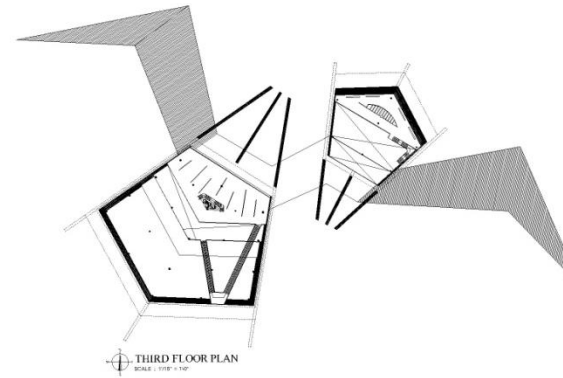
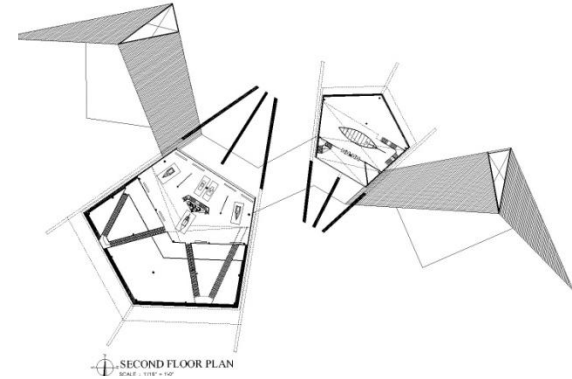
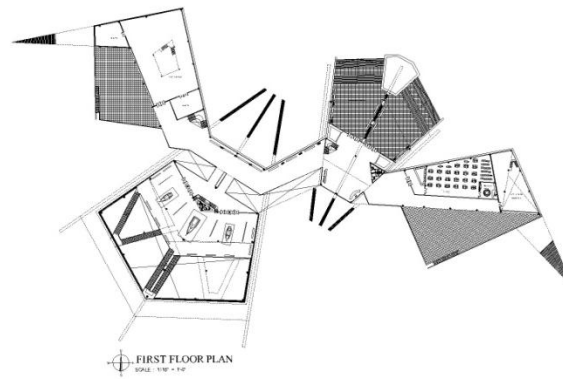
OUTDOOR SPACE



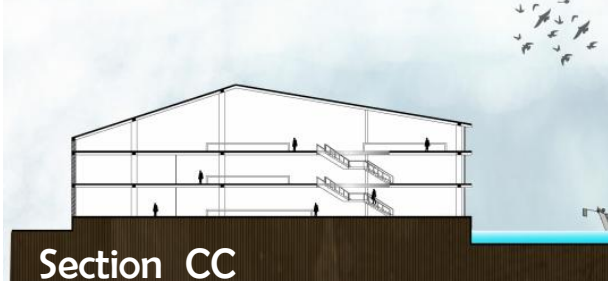
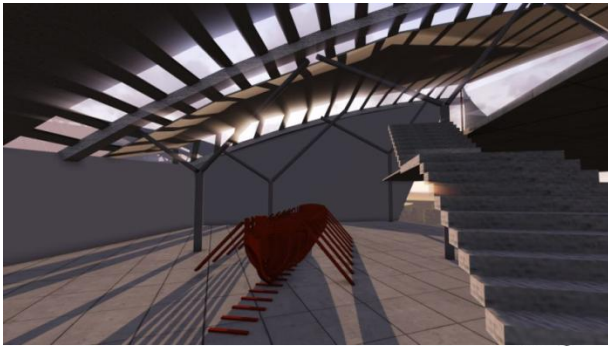
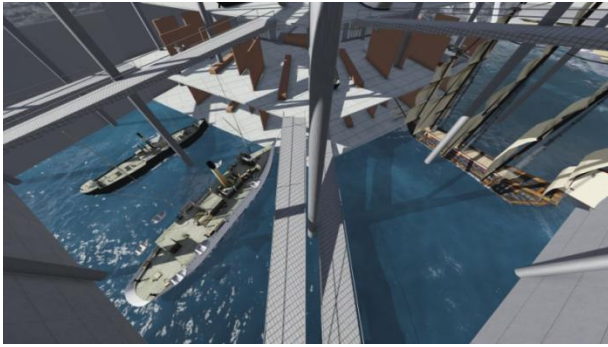
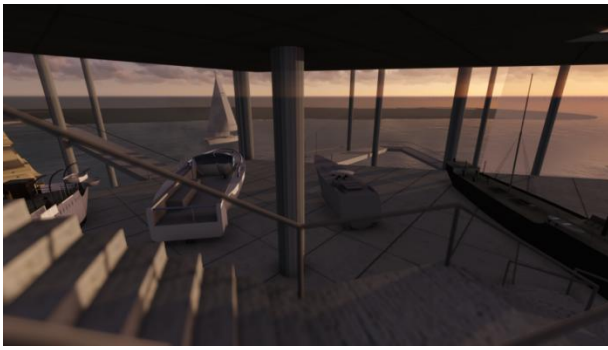
VIEW FROM SEA



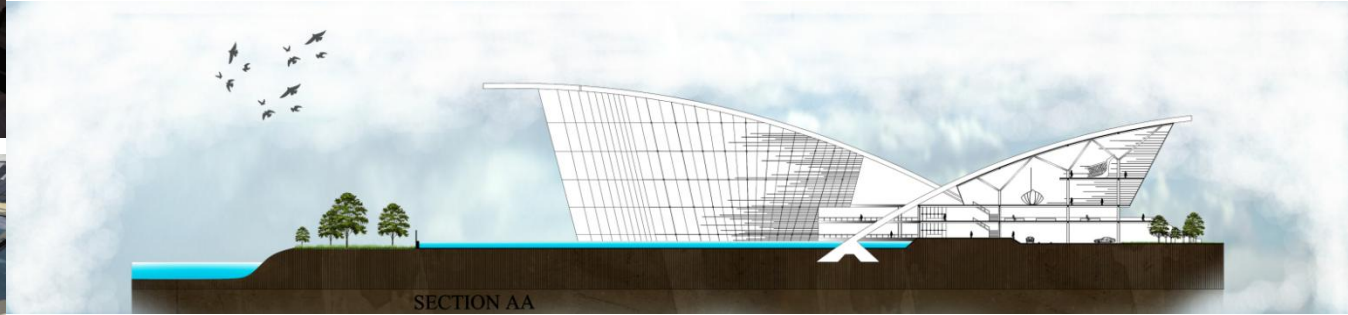
STRUCTURAL MODEL



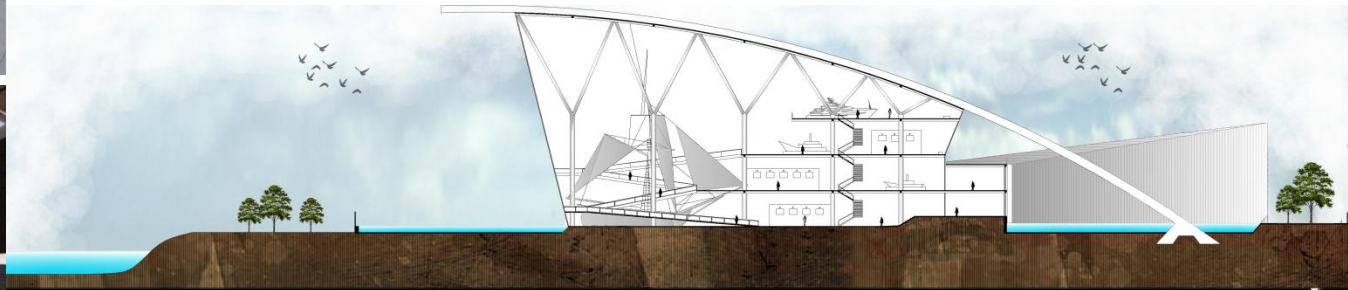
EAST ELEVATION



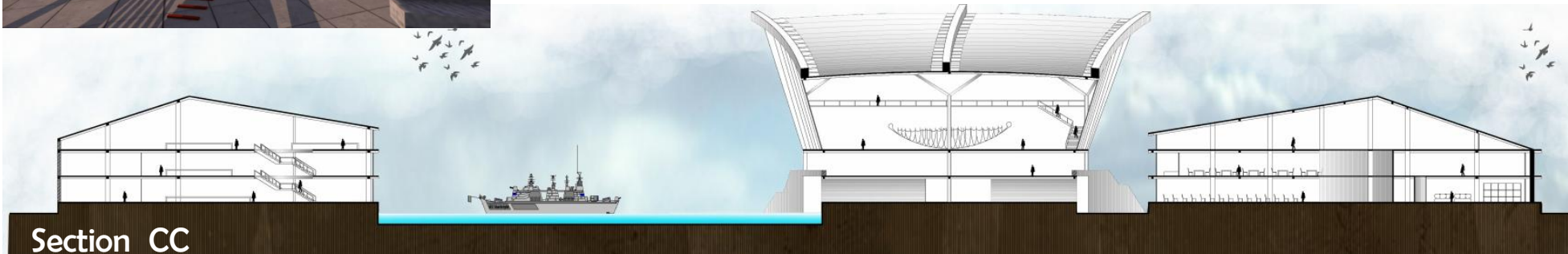
SECTIONS | RENDERS



Section AA



Section BB



Section CC

HOUSING PROJECT



Academic Project , Year : 2012

Location: Dhaka , Bangladesh

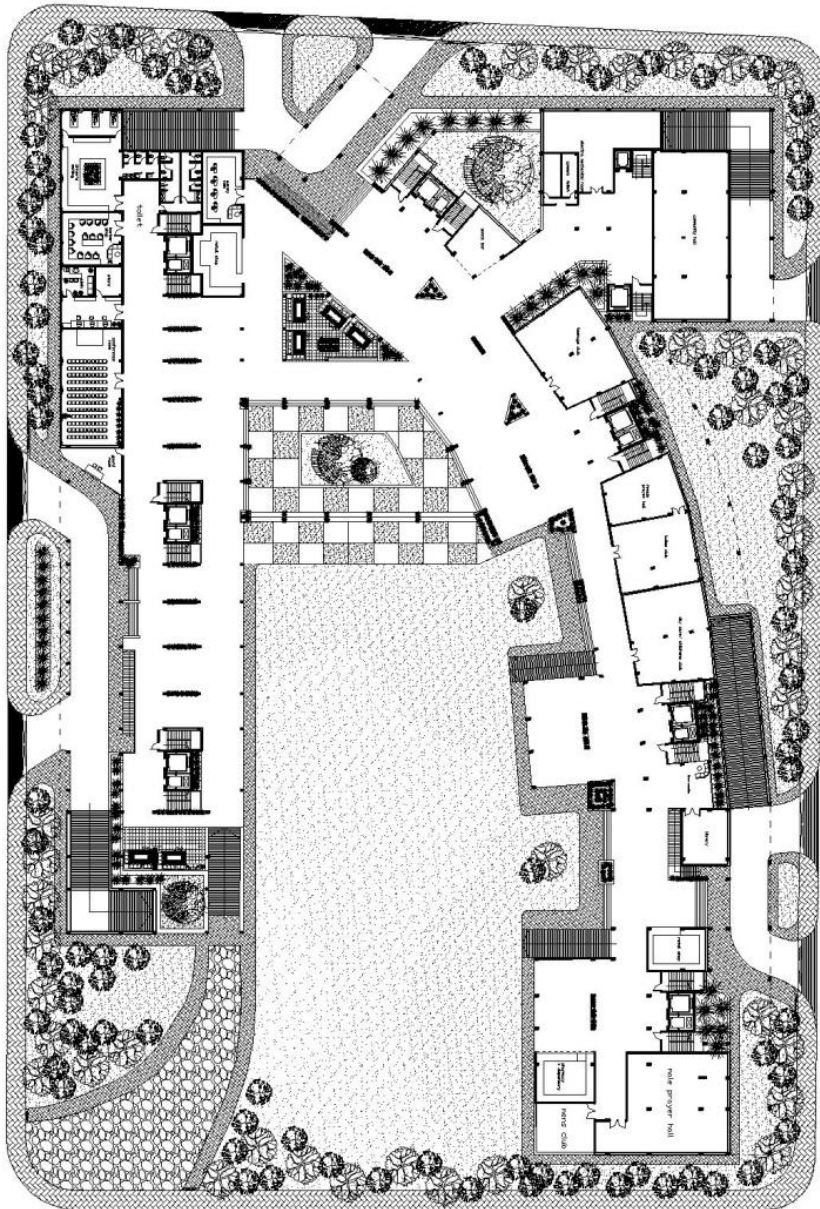
Instructor : Dr. Jasmin Ara Begum

Project type : Residential

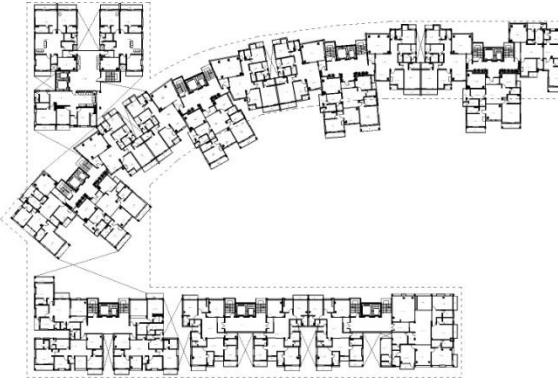
OVERVIEW

In this housing project , there is a special consideration for the elderly people in a family. In every family, we have our grand parents. Most of the families are separated from our grand parents because of our grand parents cannot get the environment of their village & also lack of their basic needs.

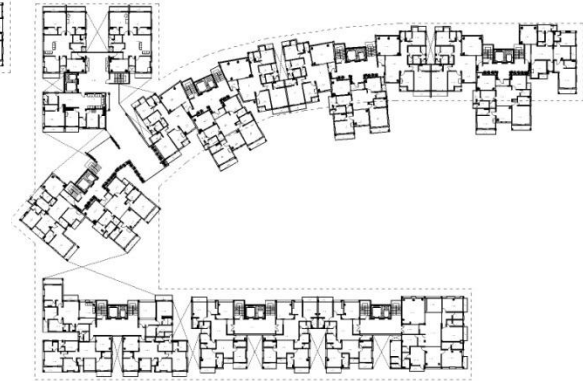
My aim was to accommodate some facilities for our grand parents in my housing project such as naturally lightened and ventilated , green courtyard and terraces everywhere , internally connected all the inhabitants , mostly the whole compound is connected with ramp for those who are unable to use the steps. This is how we can also give our new generation a proper guideline from their grand parents.



 GROUND Floor Plan



 Typical Floor Plan
(without terrace)
Scale: 1/32" = 1' - 0"



 Typical Floor Plan
(with terrace)
Scale: 1/32" = 1' - 0"





PHOTOGRAPHY

As EXTRA CURRICULAR ACTIVITY

I love doing architectural, natural photography as my hobby. Architectural Photography where art, design, and science are all. I also love travelling a lot and like to explore new places and cultures.

